MATH 1342 – Elementary Statistics – Week 7: Confidence Intervals and the Bootstrap

Oct 5 – 7 | 3.2 Understanding and Interpreting Confidence Intervals • 3.3 Constructing Bootstrap Confidence Intervals Using Standard Error | Fall 2026 · Section 019 · MW 7:30–8:50 PM · RRC Room 8210

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Construct a 95% confidence interval for a mean or proportion as statistic ± 2·SE, given the statistic and its standard error. (CLO 6) **2.** Interpret a confidence interval in context, explain the confidence level as the long-run capture rate of the method, and identify common misinterpretations. (CLO 4, CLO 6)	e-Text: 3.2 Understanding and Interpreting Confidence Intervals Slides: 3.2 Understanding and Interpreting Confidence Intervals	Participation: Confidence Intervals – which of 20 simulated CIs capture μ; fix three misinterpretations	Online Homework: 3.2 Understanding and Interpreting Confidence Intervals (due Sun Oct 11, 11:59 PM)	WileyPLUS Blackboard
3. Describe how a bootstrap sample and a bootstrap distribution are created, and explain why the bootstrap distribution is centered at the sample statistic. (CLO 5, CLO 6) **4.** Use StatKey to generate a bootstrap distribution and estimate the standard error as its standard deviation. (CLO 5, CLO 6) **5.** Construct and interpret a 95% bootstrap confidence interval using the standard-error method. (CLO 6)	e-Text: 3.3 Constructing Bootstrap Confidence Intervals Slides: 3.3 Constructing Bootstrap Confidence Intervals (embedded StatKey simulator) StatKey (lock5stat.com/statkey) – Bootstrap Confidence Intervals	Participation: Bootstrap Confidence Intervals – Wed: Try It: 32 of 50 prefer Brand A; build the bootstrap distribution, read off SE, form the CI	Online Homework: 3.3 Constructing Bootstrap Confidence Intervals (due Sun Oct 11, 11:59 PM)	WileyPLUS Blackboard StatKey
Items that cover objectives from more than one row above (objectives in parentheses)			Take-Home Quiz 4 (due Mon Oct 12, 11:59 PM) (all objectives)	Assignment Tool in Blackboard
Course resources and assessments of earlier weeks (not aligned to this week's objectives)		Participation: Sampling Distribution (Kahoot!) – Mon: review of 3.1 (Week 6 objectives)	Test 1 Retake – Mon Oct 5 – Sun Oct 11 (optional; higher score counts)	Kahoot! Assessment Tool in Blackboard Respondus LockDown Browser and Monitor

CLO 4 Explain the role of probability in statistics. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 6** Describe and compute confidence intervals.